

The Guardian — App Components Overview

1. Dashboard (The Command Center)

The entry point of the app, designed with a sleek, cyber-security aesthetic to give users immediate awareness of their security posture.

- **Live Threat Map:** An interactive, cyber-themed world map visualizer that provides a dynamic feel.
- **System Status & Metrics:** Displays the current protection level, blocked threats count, and system health.
- **Quick Actions Hub:** Fast access to primary features like Smart Scan, Secure Vault, and the DNS Filter.
- **Live Terminal:** A retro-styled terminal log that outputs background app activities in real-time.

2. Secure Modules (Core Protection)

A suite of active and proactive security tools designed to protect the device, network, and data.

I. DNS Filter (Network Protection)

- **Native Android VPN Service:** Intercepts network traffic at the OS level using split-tunneling.
- **Zero-Latency Routing:** Only intercepts DNS traffic (Port 53), ensuring normal internet speeds (streaming/browsing) are entirely unaffected.
- **Malware & Tracker Sinkhole:** Automatically blocks known malicious domains by returning NXDOMAIN (fake IP) responses.

II. Smart Scan (Threat Detection)

- **Deep System Enumeration:** Scans both installed applications and local storage directories.
- **Cloud Threat Intelligence:** Computes SHA-256 hashes and verifies them against the VirusTotal API via the backend.
- **Local Heuristic Fallback:** Automatically switches to an offline heuristic risk-scoring engine if internet is unavailable or API limits are reached.

III. Permission Auditor (Privacy Management)

- a. **Manifest Analyzer:** Evaluates all installed third-party apps for the permissions they request (Camera, Microphone, Location, etc.).
- b. **Risk Categorization:** Flags apps as Low, Medium, or High risk based on their permission footprint to help users identify invasive apps.

IV. Secure Shredder (Data Sanitization)

- a. **Custom File Explorer:** A bespoke, native file browser that bypasses standard Android UI restrictions to target exact files.
- b. **Military-Grade Obliteration:** Utilizes a native 3-pass wipe algorithm (Zeros -> Ones -> Random Entropy) via direct memory I/O, ensuring deleted files cannot be recovered by digital forensics.

V. Parental Controls (Access Management)

- a. **Domain Restrictor:** Integrates with the DNS Filter to let parents add custom websites to the blocklist (e.g., adult content or social media).
- b. **App Locker:** Runs a lightweight background thread using Android's UsageStatsManager. If a restricted app is launched, it instantly triggers a native "Access Denied" overlay, blocking the app.

3. Threat Intelligence & Education

Features designed to keep the user informed and educated about the evolving cybersecurity landscape.

I. Intel Feed (Live News)

- **RSS Aggregator:** Pulls live cybersecurity news, zero-day alerts, and threat advisories from top sources (Krebs on Security, The Hacker News, Dark Reading).
- **Unified Cyber UI:** Formats disparate news feeds into a sleek, readable, hacker-themed card layout.

II. Protection Guides & Chonjo (Training)

- **Step-by-step Modules:** Interactive tutorials teaching best practices (e.g., Phishing Defense, Password Hygiene).
- **Gamified Quizzes (Chonjo):** A level-based mini-game that tests and reinforces the user's cybersecurity knowledge.

4. Secure Vault

- **Encrypted Storage:** A hidden sector within the app that encrypts and securely stores sensitive user files, protecting them from other apps and unauthorized physical access.

5. Python FastAPI Backend (Cloud Infrastructure)

The server-side infrastructure that supports the mobile application securely.

- **API Gateway:** Handles routing and rate-limiting for mobile clients.
- **Threat Proxy (VirusTotal):** Securely manages enterprise API keys and queries the VirusTotal database, ensuring secrets are never exposed inside the mobile APK.

- **Feed Parser:** Asynchronously fetches, parses, and cleans RSS feeds from various news outlets before delivering a standardized JSON payload to the app.